

CHAPTER 1 — INFORMATION TECHNOLOGY AND SOCIETY

Question Bank with Answers • Book-style formats

1-1 Development of Information Technology and Social Transformation

1.1 Read the following passage and fill in the blanks (a)–(c): Computers were invented in the 1940s and were initially used mainly for military and scientific computation. Later, in the 1970s and 1980s, (a) became widespread, allowing individuals to use computers. In the 1990s, (b) was commercialized, advancing the globalization of information. In the 2000s, (c) emerged, leading to the explosive spread of mobile Internet.

Answer: (a) Personal computers (PCs) • (b) The Internet / the Web • (c) Smartphones

1.2 What technology, which spread from the 2010s onward, supports large-scale data analysis and the use of AI?

Answer: Cloud computing

1.3 From the following options (A–F), choose all that are emerging technologies.

- A Email
- B Autonomous driving
- C Personal computers
- D AR
- E VR
- F Quantum computing

Answer: B, D, E, F

1.4 From the following options (A–D), choose the one that most appropriately describes the characteristics of SNS.

- A A service that allows users to connect with each other and post and share information, and is highly effective at spreading information rapidly.
- B A service for buying and selling goods and services through the Internet.
- C A working style in which one works from home or other remote locations.
- D A system for making payments using electronic money or QR codes without using cash.

Answer: A

1.5 From the following options (A–D), choose the one that lists the stages of IT development in the correct chronological order.

A Birth of the computer → Rise of smartphones → Commercialization of the Internet → Spread of cloud computing

B Birth of the computer → Commercialization of the Internet → Rise of smartphones → Spread of cloud computing

C Commercialization of the Internet → Birth of the computer → Spread of cloud computing → Rise of smartphones

D Rise of smartphones → Commercialization of the Internet → Birth of the computer → Spread of cloud computing

Answer: B

1.6 Mark each statement (A–D) with ○ if true or × if false.

A Moore's Law is the empirical observation that the number of transistors on an integrated circuit doubles approximately every two years.

B Moore's Law has been said to be approaching a physical limit in recent years.

C SNS is highly effective at spreading information rapidly.

D E-commerce (EC) refers to purchasing goods at physical stores using cash.

Answer: A ○ • B ○ • C ○ • D ×

1.7 Match each description (a–c) with the most appropriate technology: A Autonomous driving, B AR, C VR.

a A technology that overlays digital information on real-world images.

b A technology that uses AI to drive a vehicle without human operation.

c A technology that allows users to immerse themselves in a virtual space generated by a computer.

Answer: a → B • b → A • c → C

1.8 What is the name of the empirical observation that the number of transistors on an integrated circuit doubles approximately every two years?

Answer: Moore's Law

1.9 What is the term for the buying and selling of goods and services using the Internet?

Answer: E-commerce (EC)

1.10 What is the term for the working style in which one works from home or other remote locations using the Internet?

Answer: Remote work

1.11 What is the term for the system for making payments using electronic money, QR codes, etc., without using cash?

Answer: Cashless payment

1.12 What is the term for the technology that uses AI to drive a vehicle without human operation?

Answer: Autonomous driving

1.13 For each of a–d, indicate which change is most closely related: A Changes in daily life, B Changes in industry and the economy, C Changes in healthcare and education.

a Taking classes through online learning.

b Paying for purchases using a mobile payment app on a smartphone.

c Sharing photos with friends on SNS.

d A company introducing a work-from-home system.

Answer: a → C • b → A • c → A • d → B

1.14 From the following options (A–D), choose the one that is NOT an appropriate description of an emerging technology.

A Autonomous driving uses AI to drive a vehicle without human operation.

B AR is a technology that overlays digital information on real-world images.

C VR is a technology that dramatically improves the processing speed of a computer.

D Quantum computing is expected to speed up computations that are difficult for traditional computers.

Answer: C

1.15 Analyze how the spread of cloud computing (from the 2010s onward) has changed the way information technology is used. [6] In your answer, refer to: large-scale data analysis, AI, and “IT as a service.”

Answer: Cloud computing delivers IT as a service over the Internet. It supports large-scale data analysis and AI by providing computing resources as a service.

1.16 If a fully cashless society were realized, choose one advantage and one possible concern, and briefly explain the reason for each.

Answer: Example: Advantage — faster and more convenient payment. Concern — people without a card or smartphone may have difficulty paying.

1.17 In autonomous driving, why is it necessary to process data instantly on the vehicle side using edge computing, rather than sending the data to the cloud for judgment?

Answer: Because even a small delay can lead to an accident. Edge computing processes data on the vehicle itself for faster decisions.

1.18 A village that has never had Internet access is connected to high-speed Internet and cashless payment for the first time. Predict two ways daily life will change, and identify one new problem.

Answer: Example: More online services and cashless payments. A possible problem is exclusion of people without suitable devices, accounts, or digital skills.

1.19 Should a community move toward cashless payment? Recommend one step and give two reasons.

Answer: Example: Move gradually while keeping cash available. This keeps the convenience of cashless payment while reducing exclusion.

1.20 Choose one emerging technology from this lesson (autonomous driving, AR, VR, or quantum computing). Suggest one way it could help solve a real problem, and state one risk.

Answer: Example: Autonomous driving could reduce driver workload; a risk is an incorrect decision caused by system or sensor problems.

1-2 How AI Works

2.1 Read the following passage and fill in the blanks (a)–(d): The general term for technologies that reproduce or perform intelligent human behavior on a computer is (a). One learning technology that makes (a) work by learning patterns from data is (b). Within (b), an advanced technology that uses neural networks is (c). Furthermore, the technology that uses (c) to generate new data is (d).

Answer: (a) AI • (b) Machine learning • (c) Deep learning • (d) Generative AI

2.2 Mark each statement (A–D) with ○ if true or × if false.

A Machine learning is one of the technologies that makes AI work.

B Deep learning is a completely different technology from machine learning.

C Generative AI is a technology that uses deep learning to generate new data.

D AI and machine learning have the same meaning.

Answer: A ○ • B × • C ○ • D ×

2.3 Match each description (a–c) with the most appropriate term: A Machine learning, B Deep learning, C Generative AI.

a A technology that learns patterns from data to make predictions and judgments.

b A technology that uses neural networks to learn complex patterns.

c AI technology that generates new data such as text and images.

Answer: a → A • b → B • c → C

2.4 What is the general term, expressed by a two-letter abbreviation, for technologies that reproduce or perform intelligent human behavior on a computer?

Answer: AI (Artificial Intelligence)

2.5 What is the term for the learning technology that makes AI work by learning patterns from data to make predictions and judgments?

Answer: Machine learning

2.6 What is the term for the advanced technology within machine learning that uses neural networks?

Answer: Deep learning

2.7 What is the term for the AI technology that uses deep learning to generate new data?

Answer: Generative AI

2.8 For each example (a–d), indicate which is most closely related: A Machine learning, B Deep learning, C Generative AI.

- a Automatically sorting spam emails.
- b Recognizing pedestrians on the road in autonomous driving.
- c Automatically generating new images from text.
- d Recommending products based on a user's purchase history.

Answer: a → A • b → B • c → C • d → A

2.9 From the following options (A–D), choose the one that correctly describes the relationship between AI, machine learning, deep learning, and generative AI.

- A AI > Deep Learning > Machine Learning > Generative AI
- B Machine Learning > AI > Generative AI > Deep Learning
- C AI > Machine Learning > Deep Learning > Generative AI
- D Generative AI > Deep Learning > Machine Learning > AI

Answer: C

2.10 From the following options (A–D), choose the one that is NOT an appropriate example of generative AI.

- A ChatGPT
- B Image generation AI
- C Spam filter
- D Audio generation AI

Answer: C

2.11 Generative AI can produce text that sounds plausible but is factually incorrect (a “hallucination”). Explain why it is dangerous to use such output as the answer to a school report as-is, and state what learners should do instead.

Answer: It may contain incorrect information that sounds convincing. Learners should verify important information using reliable sources before using it.

2.12 Analyze the relationship between AI, machine learning, deep learning, and generative AI. [6] In your answer: use the idea of nested categories, and give one example of each.

Answer: AI is the broad field; machine learning is within AI; deep learning is within machine learning; generative AI uses deep learning to generate new data. Examples can be recommendation systems, image recognition, and text/image generation.

2.13 A farmer wants to use AI to distinguish healthy crops from diseased ones in photographs. Using what you learned about deep learning and image analysis, explain what the AI would need to learn from, and identify one reason its judgment might be wrong.

Answer: It would need training examples of healthy and diseased crops. Its judgment could be wrong if the

training data is biased, insufficient, or not representative.

2.14 Suggest one helpful use and one possible misuse of generative AI, and say how the misuse could be reduced.

Answer: Example: Helpful use — generating ideas. Misuse — using false generated information as fact. Reduce it by checking facts against reliable sources.

1-3 AI in Daily Life and Industry

3.1 For each of a–d, match the AI technology used in the service with the most appropriate term: A Recommendation system, B Voice assistant, C Machine translation, D Face recognition.

a YouTube recommends videos that match the user's preferences.

b Speaking to a smartphone to check the weather.

c Translating a foreign-language website into one's native language.

d A smartphone camera automatically detects a person's face.

Answer: a → A • b → B • c → C • d → D

3.2 Mark each statement (A–D) with ○ if true or × if false.

A AI is good at probabilistic reasoning and prediction based on data.

B AI is good at recognizing images and audio.

C It is fine to leave ethical judgments entirely to AI.

D AI judgments can become inaccurate when training data is biased.

Answer: A ○ • B ○ • C × • D ○

3.3 What is the term for the system that predicts preferences from past behavior data and displays recommendations?

Answer: Recommendation system

3.4 What is the term for the AI technology that recognizes a voice, understands commands, and executes them?

Answer: Voice assistant

3.5 In which industry are image-diagnosis AIs used?

Answer: Healthcare

3.6 In manufacturing, what is the term for the system that predicts product failures in advance?

Answer: Predictive maintenance

3.7 From the following options (A–D), choose the one that is NOT an appropriate description of services that use AI.

A A recommendation system predicts preferences from past behavior data and displays recommendations.

B A voice assistant recognizes a voice, understands commands, and executes them.

C Machine translation is a system in which humans manually translate foreign languages.

D Face recognition automatically detects and identifies human faces in photographs.

Answer: C

3.8 From the following options (A–D), choose all that require caution when using AI.

- A High-speed processing of large amounts of data
- B Ethical judgments
- C Judgments involving personal information or privacy
- D Recognition of images and audio

Answer: B and C

3.9 Select all options that AI excels at.

- A Ethical judgments
- B Finding features in images and text and classifying them
- C Judgments involving personal information
- D Probabilistic reasoning and prediction based on data
- E Decision-making that bears responsibility for results
- F Recognition and generation of images, audio, and text

Answer: B, D, F

3.10 For each AI usage example (a–d), indicate which industry is most closely related: A Healthcare, B Agriculture, C Manufacturing, D Logistics.

- a Optimizing delivery routes.
- b Detecting diseases from X-ray images.
- c Detecting pests and diseases.
- d Automating product quality inspection.

Answer: a → D • b → A • c → B • d → C

3.11 Analyze why AI is well suited to some tasks but requires caution in others. [6] In your answer: refer to what AI is good at (finding patterns, recognition, prediction) and to what needs caution (ethical judgments, privacy, final decision-making).

Answer: AI is strong at finding patterns, recognition, generation, and prediction. Caution is needed for ethical judgments, privacy, final decisions, and responsibility because AI can be biased or wrong.

3.12 In agriculture, give one realistic advantage of introducing AI and one concern that should be considered before introduction, and briefly explain the reason for each.

Answer: Example: AI can detect pests or diseases early. A concern is incorrect judgment from poor or insufficient data.

3.13 A hospital is considering using image-diagnosis AI to detect diseases from X-ray images. Explain why the final diagnosis should be confirmed by a human doctor rather than left to AI alone.

Answer: AI can identify image patterns but can be wrong. A doctor should check the result, consider the patient, and make the final responsible diagnosis.

3.14 AI services often use your personal data — what you watch, buy, or say. Why does this raise privacy concerns, and who should decide how your data is used?

Answer: Because personal information and behavior can be collected and analyzed. Its use should be controlled by appropriate rules that protect privacy.

3.15 A school library wants to use AI to recommend books to students based on what they have borrowed before. Explain how the AI would work, and identify one caution the school should consider.

Answer: It could use borrowing history to predict preferences and recommend books. A key caution is protecting students' personal data.

1-4 Ethical Issues with AI

4.1 Mark each statement (A–D) with ○ if true or × if false.

A Algorithmic bias is bias in AI judgments caused by bias in the training data.

B Even if an AI's decision-making process is opaque, there is no problem as long as the result is correct.

C Explainable AI (XAI) is a technology that makes it possible for humans to understand the AI's reasoning.

D When an AI makes an incorrect judgment, who is responsible has already been clearly determined.

Answer: A ○ • B × • C ○ • D ×

4.2 Match each description (a–d) with the most appropriate AI ethics basic principle: A Fairness, B Transparency, C Privacy protection, D Accountability.

a Ensuring that a hiring AI evaluates fairly regardless of gender.

b Disclosing the AI's decision-making process to users in an easy-to-understand way.

c Not using collected personal data for purposes other than as originally intended.

d Being answerable when an AI's diagnostic result turns out to be wrong.

Answer: a → A • b → B • c → C • d → D

4.3 What is the term for the phenomenon in which bias arises in AI judgments due to bias in the training data?

Answer: Algorithmic bias

4.4 What is the term for the technology that makes it possible for humans to understand why an AI made a particular judgment? Also, give its three-letter abbreviation.

Answer: Explainable AI (XAI)

4.5 Among the basic principles of AI ethics, what is the term for the principle that requires not unjustly discriminating against any particular person or group?

Answer: Fairness

4.6 Among the basic principles of AI ethics, what is the term for the principle that requires handling personal information appropriately?

Answer: Privacy protection

4.7 From the following options (A–D), choose the one that is NOT an appropriate cause of algorithmic bias.

A Training data is biased.

B Past discriminatory tendencies are reflected in the data.

- C The processing speed of the computer is slow.
- D There is insufficient data on certain attributes.

Answer: C

4.8 From the following options (A–D), choose all that are privacy issues associated with AI.

- A Individuals are identified and tracked through face-recognition technology.
- B Online behavior data is collected and analyzed in large quantities.
- C AI creates creative works.
- D AI computation speed improves.

Answer: A and B

4.9 From the following options (A–D), choose the one that most appropriately describes ethical issues with AI.

- A There are no ethical issues with AI.
- B Algorithmic bias is caused by the AI's processing speed.
- C Bias in training data can cause discriminatory bias in AI judgments.
- D Even if an AI's decision-making process is opaque, there is no problem for users.

Answer: C

4.10 Analyze how algorithmic bias can arise in an AI system, and how it can be reduced. [6] In your answer: refer to the cause of bias (biased or insufficient training data) and to how the principles of fairness and transparency help address it.

Answer: Bias can arise when training data is biased, reflects past discrimination, or is insufficient for some groups. It can be reduced by checking/improving data, testing results across groups, and applying fairness and transparency.

4.11 A hiring AI was found to rate applicants of a particular gender lower. Using the ideas of algorithmic bias and the basic principles of AI ethics, explain the most likely cause and one possible measure to address it.

Answer: A likely cause is biased or insufficient training data. A measure is to check the data and test the AI across different groups, using fairness and transparency.

4.12 When an AI makes an unfair decision, who is responsible — the developer, the company using it, or the operator? Note the responsibility each one holds.

Answer: Responsibility may involve the developer, the company, and/or the operator depending on the circumstances. The textbook notes that no single clear standard has been established.

4.13 Before a company uses a hiring AI, what one rule should it follow to make the system fairer? Justify your rule using the principles of AI ethics.

Answer: Example: Require fairness testing across different groups before and during use. This supports

fairness and transparency.

4.14 A school wants to use face-recognition cameras at its gate to record attendance automatically. Identify one practical benefit and one privacy concern. Then, if the camera misidentifies a student, who should be responsible — the school, or the company that made the AI? Give a reason.

Answer: Benefit: automatic attendance. Concern: identification/tracking and use of personal data. Responsibility should be considered for both the school and the company according to who controlled the relevant decision or failure.

Mixed Chapter 1 Review

5.1 Which option gives the correct relationship among the AI concepts?

- A AI > Deep Learning > Machine Learning > Generative AI
- B Machine Learning > AI > Generative AI > Deep Learning
- C AI > Machine Learning > Deep Learning > Generative AI
- D Generative AI > Deep Learning > Machine Learning > AI

Answer: C

5.2 Which option is NOT correct?

- A E-commerce means buying and selling through the Internet.
- B AR overlays digital information on the real world.
- C VR is a technology that dramatically improves the processing speed of a computer.
- D Cloud computing can support large-scale data analysis and AI.

Answer: C

5.3 Which set contains only AI ethics principles from the chapter?

- A Fairness, transparency, privacy protection, accountability
- B Speed, storage, privacy, networking
- C Accuracy, bandwidth, encryption, fairness
- D Transparency, e-commerce, accountability, VR

Answer: A

5.4 Explain in one sentence why information technology development is not only about faster machines.

Answer: Each stage of IT also changed how society communicates, works, learns, pays, and handles information.

5.5 Explain in one sentence why human judgment is still needed when using AI.

Answer: AI is powerful at patterns and prediction, but people are still needed for ethics, privacy, accuracy checking, final decisions, and responsibility.

5.6 Write the four basic principles of AI ethics.

Answer: Fairness • Transparency • Privacy protection • Accountability

5.7 Write the major stages of IT development presented in the chapter.

Answer: Birth of the computer → spread of personal computers → commercialization of the Internet/Web →

rise of smartphones → spread of cloud computing.

5.8 A student asks an AI to write a report. The AI gives a confident but incorrect fact. What should the student do?

Answer: Verify the information using reliable sources and correct or remove unsupported information before using the output.